

Where the Problem Stands: Contextual States on Concrete Orthomodular Lattices

A hand-held mathematical exposition

How to read this document. Every mathematical assertion below is tagged with its epistemic status, using the following convention. A **Theorem**, **Proposition**, **Corollary**, or **Lemma** is proved in this programme’s own work; a proof sketch is given, honestly marked as compressed where it is. An **External Theorem** is proved in the published literature and is cited, not re-derived. A **Conjecture** is a precise open question, stated but not asserted. A **Heuristic** is an intuition that has guided the search but has no proof behind it — flagged explicitly as such. A **Falsified** claim is one that was seriously investigated and shown not to hold; it is recorded so that a dead idea is never mistaken for a live one. Nothing below is stated more strongly than this convention allows.

Summary

This document explains where a specific open problem in the foundations of quantum logic currently stands. The problem asks: can a *concrete* system of yes/no questions — one built honestly from subsets of an underlying sample space, with no exotic algebra — support a probability assignment that is consistent at every finite stage of questioning, yet provably cannot be extended to a fully countably-additive probability measure? Such an assignment would be a genuinely *contextual* probability: locally coherent, globally impossible to complete. The answer depends on how rigid the underlying logical structure is required to be. If the structure is only required to be an *orthomodular poset* — a mild associativity requirement, weaker than a lattice — then the answer is **yes**, and this is a proved theorem of Zermelo–Fraenkel set theory with choice (ZFC): an explicit example exists. If the structure is required to be a full *orthomodular lattice* — every pair of questions must have a well-defined greatest common refinement — then the answer is **conjectured to be no**, and this conjecture is currently open. Three independent finite obstruction results, together with a recently proved external theorem about a closely related but different mathematical object, all point the same way: toward the conjecture being true. But none of them proves it. The exact reason a proof is hard is explained in Section 5: the obstruction that would be needed is a genuinely infinite combinatorial phenomenon (living at the level of the first uncountable ordinal, ω_1), and every finite method that has been tried provably cannot reach that far. Two precise, still-open mathematical routes toward closing the gap are stated in Section 6.

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1 The setting, from scratch

We are going to build up, slowly, the mathematical objects the problem is about. Nothing here requires familiarity with quantum logic; it requires only comfort with Boolean algebras, measures, and basic order theory. Every definition gets a one-sentence plain-language gloss before its formal statement, and a small worked example.

1.1 Why orthomodular lattices, informally

Think of a system you can ask yes/no questions about — a physical system, a database, an experiment. Some questions are *compatible*: you can ask them together and get a joint answer (“is the coin heads, and is it Tuesday?”). Classical probability theory assumes *all* questions are compatible with all others: the collection of possible questions/events forms a Boolean algebra. Quantum mechanics is the historical motivation for dropping this assumption. Some pairs of questions (position and

momentum, say) cannot both be asked with a definite joint answer. The algebraic structure that survives when you keep the order-theoretic skeleton of “event A implies event B ” and “the complementary event of A ” but drop the assumption that any two events have a joint refinement is an *orthomodular lattice*. This document is entirely about a purely mathematical question concerning such structures; no physics is assumed or needed.

1.2 Orthomodular lattices

Definition 1.1 (Orthomodular lattice). ***Plain gloss:** a lattice of “events” with a complementation operation (“the event did not happen”) that behaves sensibly, except that we do not require it to satisfy the distributive law that makes an algebra Boolean.*

Formally, an orthomodular lattice is a bounded lattice L (with least element 0 and greatest element 1 , and meet/join operations $\wedge, \vee$) equipped with an order-reversing involution $a \mapsto a^\perp$ (the orthocomplement) such that:

- (i) $a \vee a^\perp = 1$ and $a \wedge a^\perp = 0$ for every a ;
- (ii) $a \leq b \implies b^\perp \leq a^\perp$;
- (iii) $a^{\perp\perp} = a$;
- (iv) (the orthomodular law) if $a \leq b$ then $b = a \vee (b \wedge a^\perp)$.

Two elements a, b are compatible (or orthogonal if in addition $a \leq b^\perp$) according to further standard definitions we will not need in general form. A Boolean algebra is exactly an orthomodular lattice in which every two elements are compatible (the distributive law holds); orthomodular lattices that fail this are the genuinely non-classical case.

Example 1.2 (The smallest non-Boolean example: MO_2). Take two disjoint “two-outcome” experiments and glue them at their top and bottom. Concretely: the six-element lattice with least element 0 , greatest element 1 , and two incomparable pairs of atoms/coatoms $\{a, a^\perp\}$ and $\{b, b^\perp\}$, where $a \wedge b = 0$ but a and b are *not* orthogonal in the algebra (there is no third event that makes $a \vee b$ decompose the way a Boolean union would). This is the smallest orthomodular lattice that is not a Boolean algebra: it satisfies the orthomodular law but fails distributivity. It will recur below as the standard test object for “does the failure survive being tiny and finite,” and the answer for MO_2 (and every finite orthomodular lattice) is always *no* — see Proposition 3.6 below.

Example 1.3 (A slightly bigger picture: the pentagon). A second standard small example, often drawn as a five-cycle of atoms (a *Greechie diagram*: nodes are atoms, edges join atoms that generate a common Boolean block), is the *pentagon* lattice: five atoms arranged around a cycle, each pair of *adjacent* atoms compatible (living in a shared small Boolean piece), but non-adjacent pairs incompatible. Like MO_2 , it is a finite, concrete, non-Boolean orthomodular lattice, and — this will matter later, in Section 7 — it is a standard test bed for checking whether a proposed structural mechanism is really doing the work claimed for it, precisely *because* it is finite and hence automatically tame in the sense defined below.

1.3 Concrete (set-representable) orthomodular lattices

Everything in this document concerns *concrete* orthomodular lattices: ones built honestly out of actual subsets of an actual set, with no abstract algebra hiding underneath.

Definition 1.4 (Concrete σ -complete orthomodular lattice). ***Plain gloss:** a family of subsets of a set Ω that is closed under complementation and under taking countable unions of disjoint members, and in which every two members have a genuine set-theoretic greatest common refinement inside the family.*

Formally, fix a set Ω (the sample space). A concrete σ -class on Ω is a family $L \subseteq \mathcal{P}(\Omega)$ containing $\emptyset$ and Ω , closed under set-complement, and closed under countable unions of pairwise-disjoint members of L . The order on L is set-inclusion, and the orthocomplement is set-complement, $E^\perp = \Omega \setminus E$. Every concrete σ -class is automatically an orthomodular poset: a partially ordered set with orthocomplementation satisfying the orthomodular law, but where two elements need not have a well-defined meet inside the family. If, further, every two elements of L possess a $\subseteq$ -greatest common lower bound that is itself a member of L (a genuine lattice meet), we call L a concrete σ -complete orthomodular lattice. The distinction — poset versus lattice — is exactly the boundary this entire document is about, and it will be named **OMP** (orthomodular poset — meets not required to exist) versus **OML** (orthomodular lattice — meets required to exist) from here on.

Concreteness matters. An abstract orthomodular lattice can always be represented, in principle, in some more exotic way; a *concrete* one is tied down to honest subsets of a set, so its truth-values are honest membership facts. This is the setting in which the projection lattice of a Hilbert space, for instance, generically *fails* to be concrete (Kochen–Specker, discussed in passing below) — but the objects in this document are always concrete by definition, so that obstruction is simply not in play here.

1.4 States: finitely additive and σ -additive

Definition 1.5 (Two-valued state). ***Plain gloss:** an assignment of “true”/“false” to a family of events that behaves like a genuine yes/no probability: additive over finite disjoint unions, and (in the stronger version) additive over countable disjoint unions too.*

Let B be a $\perp$ -closed subfamily of L (i.e. closed under complementation; B need not be all of L). A two-valued state on B is a function $s : B \rightarrow \{0, 1\}$ with $s(\Omega) = 1$, such that for every finite pairwise-disjoint family $\{A_i\} \subseteq B$ whose union lies in B ,

$$s\left(\bigcup_i A_i\right) = \sum_i s(A_i).$$

s is σ -additive if the same identity holds for every countable pairwise-disjoint family. A point-evaluation $\delta_\omega(E) := \mathbf{1}[\omega \in E]$ for a fixed $\omega \in \Omega$ is always a σ -additive state; such states are called Dirac. Every other state is non-Dirac.

Intuitively: a finitely additive state is a consistent finite-stage probability assignment. A σ -additive one survives being pushed to the countable limit — the assignment to a countable disjoint union really is the sum of the assignments to the pieces, with no slippage. The gap between these two notions is exactly where the whole problem lives.

1.5 Blocks (maximal Boolean subalgebras)

Definition 1.6 (Block). ***Plain gloss:** the largest “classical” piece of the structure containing a given collection of events — the maximal set of mutually compatible events, which on its own behaves exactly like an ordinary Boolean algebra. A block of L is a maximal subfamily $M \subseteq L$ that is pairwise compatible (every two members of M generate a Boolean subalgebra inside L).*

Every element of L lies in at least one block (a standard Zorn’s-lemma argument), and every block, viewed on its own, is an honest Boolean σ -algebra of subsets of Ω : within a single block there is no obstruction at all, and ordinary classical probability theory applies without qualification.

This will be the single most important structural fact used below: *all* non-classicality lives strictly *between* blocks, never inside one. A countable disjoint family is always tested for σ -additivity inside a single block (Theorem 3.1 below makes this precise), so the failure of σ -additivity we are chasing is a failure of *assembling* compatible pieces across blocks, not a failure inside any one block.

1.6 Essential contextual states: the object we are looking for

Definition 1.7 (Essential contextual state). ***Plain gloss:** a probability assignment that is completely reasonable if you only ever ask finitely many questions at a time, but which no amount of countably-additive bookkeeping can ever complete — not because of some pathological infinite question, but because the finite data itself is already incompatible with every possible σ -additive completion.*

Fix a concrete σ -class L on Ω , a finite $\perp$ -closed sub-poset $B \subseteq L$, and a two-valued state s_0 on B . Say s_0 is finitely coherent on L if s_0 extends to some finitely additive two-valued state defined on all of L (i.e. s_0 is not locally self-contradictory; it is a legitimate finite fragment of some global finitely additive assignment). Say s_0 is a σ -essential contextual state if s_0 is finitely coherent on L , but no σ -additive two-valued state on all of L extends it.

The finite-coherence clause is not decorative. Without it, the definition would be trivially satisfiable by a self-contradictory finite fragment that no state of *any* kind extends — an uninteresting failure with nothing to do with the countable-additivity gap. Section 3 makes this precise (Proposition 3.2); for now, hold onto the plain-language version: an essential contextual witness is a piece of probability data that is honestly extendable to *some* weak (finitely additive) global assignment, yet is provably incompatible with *every* strong (countably additive) one.

Definition 1.8 (Tame). *We say a concrete σ -class L is tame (or satisfies finite-trace σ -liftability, written $\Phi(L)$) if every finitely coherent finite pattern on L extends to some σ -additive two-valued state on L — i.e. L admits no σ -essential contextual state at all. Tameness is the negation of “carries a witness”: L is tame exactly when the search of this document, run on L , comes back empty.*

2 The central question, stated

We can now state the problem precisely.

Conjecture 2.1 (The central conjecture). *Every concrete σ -complete orthomodular lattice (every two elements have a genuine meet in the structure) is tame: it admits no σ -essential contextual state.*

This is the headline question. It is currently **open**. What is known, precisely, is the following dichotomy, which this whole document exists to explain:

- If L is only required to be a concrete σ -complete orthomodular *poset* (Definition 1.4, without the lattice/meet requirement), then Conjecture 2.1’s analogue is **false**: an explicit witness is proved to exist, in plain ZFC, with no extra set-theoretic hypotheses (Theorem 4.1 below).

- Conjecture 2.1 itself, for genuine *lattices*, remains open. The gap between these two facts — poset versus lattice — is the entire content of what remains to be understood, and it is discussed at length in Sections 4–6.

2.1 A second, equivalent, face of the conjecture

The conjecture has one genuinely useful reformulation that this programme has proved equivalent to it (not merely suggestively related to it), and we record it now because later sections work with whichever face is more convenient.

Theorem 2.2 (Cluster reformulation of tameness). *Fix a concrete σ -class L . Call a finite family $\mathcal{C} \subseteq L$ an f.a.-coherent cluster if some finitely additive two-valued state on L assigns value 1 to every member of $\mathcal{C}$. Then the following are equivalent:*

- (a) (Tameness, block form) *L is tame in the sense of Definition 1.8: every finite pattern on a block B that agrees with some finitely additive state on L extends to a σ -additive state on L .*
- (b) (Tameness, cluster form) *Every f.a.-coherent cluster $\mathcal{C}$ is assigned the value 1, throughout, by some single σ -additive two-valued state on L .*

Proof sketch. Both directions are elementary manipulations that use no special structure of L (in particular, no latticehood): given a block-level pattern, close it under complementation and read off its value-1 part as a cluster; conversely, given a cluster, extend it to a full pattern on the $\perp$ -closure of the cluster by declaring the complements false, and check the state constructed for the cluster form already satisfies every complement constraint automatically. Full formal detail is routine bookkeeping and has been machine-verified in both directions. $\square$

Plain gloss. These are two ways of asking the same question. The block form says: “does every consistent finite yes/no pattern, read off one classical piece at a time, extend to a genuine global probability measure?” The cluster form says the same thing more economically: “does every finite set of events that some finitely-additive assignment can jointly mark ‘true’ get simultaneously marked ‘true’ by some *countably*-additive assignment?” We use whichever phrasing is lighter in context; they never disagree.

We do *not* present a third independent face here. A superficially tempting third reformulation — splitting the negation of tameness into “no point-evaluation realizes the pattern” and “no other σ -additive state realizes it either” — is a genuine and useful fact, but it is a logical *tautology* (every σ -additive state is either a point-evaluation or it isn’t) rather than an independent reformulation of the conjecture; we develop it in Section 3 below (Theorem 3.4), where its actual mathematical content (one of the two halves is always free, so the entire difficulty is concentrated in the other) belongs.

3 What is proved: the solid ground

We now build up, as a sequence of labelled theorems, everything that is actually established. Each is followed by one plain sentence on what it buys strategically.

3.1 All non-classicality lives between blocks

Theorem 3.1 (Blocks are Boolean σ -fields). *Every block M of a concrete σ -class L is, restricted to itself, an honest Boolean σ -algebra of subsets of Ω : it is closed under complement and under countable unions of pairwise-disjoint members, and any countable disjoint family entirely contained*

in M has its (automatically well-defined) union already inside M . Consequently, testing whether a state is σ -additive on a countable disjoint family reduces to testing it on the single block containing that family.

Proof sketch. A block is by definition a maximal pairwise-compatible subfamily; pairwise compatibility together with maximality forces closure under the Boolean operations restricted to M , by a standard Zorn’s-lemma/maximality argument common to orthomodular lattice theory. $\square$

What this buys. It licenses the entire strategy of the programme: because σ -additivity only ever has to be checked one block at a time, the search for an essential contextual state reduces to a question about how compatible finite patterns on *different* blocks can (or cannot) be simultaneously realized by a single global σ -additive state. All of the difficulty is an assembly problem, never a within-block problem.

3.2 Why the finite-coherence clause is load-bearing

Proposition 3.2 (Boolean baseline). *If L is (all of) a Boolean σ -algebra of subsets of Ω (i.e. L has only one block, itself), and s_0 is finitely coherent on L in the sense of Definition 1.7, then s_0 extends to a Dirac (hence automatically σ -additive) state on L . Consequently L carries no σ -essential contextual state at all: tameness on a Boolean carrier is not merely conjectured but immediate from the definitions.*

Proof sketch. Let $\mu \supseteq s_0$ be any global finitely additive two-valued state witnessing s_0 ’s coherence, and let B_0 be the finite Boolean subalgebra generated by B . The atoms of B_0 partition Ω into finitely many pieces; finite additivity of μ over this partition forces exactly one atom a to carry μ -value 1, and a is nonempty. Any point $\omega \in a$ then gives a Dirac state δ_ω agreeing with μ (hence with s_0) on all of B_0 , since every member of B_0 either contains a or is disjoint from it. $\square$

Why the finite-coherence clause matters. Without it, one could manufacture a spurious “witness” out of nothing: simply write down a finite pattern that is locally inconsistent even at the finite, finitely-additive level (no global finitely additive assignment agrees with it at all). Such a pattern trivially fails to extend σ -additively — but for a vacuous reason that has nothing to do with the countable-additivity gap this document is about. Requiring finite coherence rules this out: it forces the pattern to be a genuine fragment of *some* legitimate global finitely-additive assignment, so that its failure to extend all the way to a σ -additive one is a meaningful, non-trivial fact. Proposition 3.2 shows this is not merely a technical nicety: on the most classical carrier of all (Boolean), finite coherence by itself already forces the strongest possible outcome, a Dirac extension. Any genuine witness must therefore live on a carrier where finite coherence does *not* force this — i.e. a non-Boolean one, as the theorem of Section 4 in fact does.

3.3 Countably generated blocks force triviality

Theorem 3.3 (Countably generated boundary extension). *Let A be a countably generated Boolean σ -subalgebra of L (i.e. generated, as a σ -algebra, by countably many of its members), and suppose A embeds faithfully, via a σ -homomorphism, into a σ -complete orthomodular lattice L whose σ -additive two-valued states order-separate its elements (distinct nonzero elements are distinguished by some σ -state). Then every σ -additive two-valued state on A extends to one on L .*

Proof sketch. Fix generators $(a_n)_{n < \omega}$ of A and a σ -state μ on A . For each n let $b_n = a_n$ if $\mu(a_n) = 1$ and $b_n = a_n^\perp$ otherwise; the finite conjunctions $b_1 \wedge \cdots \wedge b_n$ all have μ -value 1, so countable additivity of μ forces their countable meet $c_\mu := \bigwedge_n b_n$ to be nonzero. Order-separation in L then

supplies a σ -state on L that charges $j(c_\mu)$ (the faithful image of c_μ), and this state agrees with μ on every generator, hence on all of A . $\square$

What this buys. It rules out an entire class of would-be witnesses at a stroke: if the finite pattern you are trying to make essential lives inside — or can be faithfully and separatingly embedded into — a *countably generated* piece of the structure, it will always extend. Any witness, if one exists on a lattice, cannot be countably generated in this separating sense; genuine uncountability of structure is a necessary ingredient, not an incidental feature.

3.4 The Dirac/non-Dirac partition (a structural tautology, with real content)

Theorem 3.4 (Localization to the non-Dirac core). *Fix a finite $\perp$ -closed $B \subseteq L$ and a two-valued state s_0 on B . Then s_0 extends to no σ -additive two-valued state on L if and only if both of the following hold:*

- (i) *no Dirac (point-evaluation) state extends s_0 : equivalently, $\bigcap\{A \in B : s_0(A) = 1\} = \emptyset$;*
- (ii) *no non-Dirac σ -additive two-valued state extends s_0 either.*

Proof. Every σ -additive two-valued state is, by Definition 1.5, either Dirac or non-Dirac, and these two possibilities are mutually exclusive and jointly exhaustive; “no σ -state extends s_0 ” is therefore logically equivalent to “no Dirac state extends s_0 , and no non-Dirac one does either.” The restatement of clause (i) as an intersection condition is immediate: a point ω satisfies $\delta_\omega \upharpoonright B = s_0$ exactly when ω lies in every s_0 -true member of B (the s_0 -false constraints are automatic once the true ones are met, by $\perp$ -closure of B). $\square$

Why this is a tautology, and why it is still useful. The equivalence itself is definitionally forced — it says nothing more than “a state is Dirac or it isn’t” — and so it is *not*, on its own, an independent reformulation of tameness in the sense of Theorem 2.2. Its value is entirely strategic: clause (i) above (does some point realize the pattern) is a purely finite, completely checkable condition, and — as Section 4’s construction shows by explicit example — it can always be arranged to hold or fail as desired, essentially for free, independently of clause (ii). This means the *entire* difficulty of finding or ruling out a witness collapses onto clause (ii): the question of whether some *non-Dirac* σ -additive state realizes the pattern. Every substantive result in the rest of this section, and the construction of Section 4, is usefully read as being *about* clause (ii), with clause (i) treated as settled machinery.

3.5 The necessary condition on any witness, with its exact choice-strength

This is the sharpest structural fact proved to date about what a witness *must* look like, and it comes with an unusually precise calibration of how much of the axiom of choice it actually requires — worth spelling out carefully, because the calibration itself is part of the content.

Theorem 3.5 (Non-extendability, and the equivalence it sits inside). *Let s_0 be a two-valued state on a finite $\perp$ -closed $B \subseteq L$, and let $V := \{A \in B : s_0(A) = 1\}$ (the “value-1 family” of s_0). The following three statements are equivalent:*

- (1) *some point $\omega \in \Omega$ lies in every member of V (i.e. s_0 is realized by a Dirac state);*
- (2) *s_0 extends to a two-valued state on the Boolean σ -algebra generated by B ;*
- (3) *V has the finite intersection property: every finite subfamily of V has nonempty intersection.*

Directions $(1) \Rightarrow (3)$ and $(2) \Rightarrow (3)$ hold in plain ZF, with no choice at all. The remaining direction, $(3) \Rightarrow (2)$, is exactly an application of the Boolean Prime Ideal theorem (BPI) — the standard weak form of the axiom of choice equivalent to “every proper filter on a Boolean algebra extends to an ultrafilter” — and no weaker principle suffices for this step in general.

Consequently: if s_0 is a σ -essential contextual state, then (since no Dirac state realizes it, by definition of essentiality) V fails the finite intersection property at some finite stage: some finite subfamily of the value-1 family already has empty intersection. This finite-stage failure statement itself is a plain ZF consequence (via the contrapositive of $(1) \Leftrightarrow (3)$, which needs no choice) — it is the abstract three-way equivalence, not the concrete failure fact, that is calibrated exactly at BPI.

Proof sketch. $(1) \Rightarrow (3)$: if some ω lies in every member of V , every finite subfamily of V meets at ω , trivially. $(2) \Rightarrow (3)$: an extension to the generated Boolean σ -algebra, being finitely additive and two-valued there, forces finite intersections of value-1 sets to remain value-1 (hence nonempty), by the ordinary finite-additivity argument inside a Boolean algebra. $(3) \Rightarrow (2)$: the finite intersection property makes V a filter base; extend it to an ultrafilter on the generated Boolean algebra (this step is the Boolean Prime Ideal theorem, precisely) and read off the corresponding two-valued state. $\square$

What this buys, and why the choice-strength matters. Strategically: any witness search can restrict attention, without loss, to finite value-1 families that fail to have a common point at some finite, checkable stage — this is a sharp, computationally testable necessary condition, and it was checked directly against the one known witness (Theorem 4.1 below): its value-1 family fails at the smallest possible stage, three. For the reverse-math-minded reader: the fact that direction $(3) \Rightarrow (2)$ needs exactly BPI, and not more, tells you precisely how “constructive” this piece of the theory is — the *obstruction* itself (the finite-stage failure, used as a necessary condition on witnesses) is choice-free, while the *positive* extension direction genuinely needs a weak form of choice and nothing stronger. This is a clean example of a place where the reverse-mathematical strength of a classical-logic fact has been pinned down exactly rather than left vague.

3.6 The finite obstruction results

Proposition 3.6 (No finite or countably-generated witness). *No finite concrete orthomodular poset (lattice or not) carries a σ -essential contextual state: on a finite carrier, every finitely additive two-valued state is automatically σ -additive (there is nothing infinite to fail to converge on), so tameness is automatic. More specifically, the following assembly obstruction holds: no concrete σ -complete orthomodular lattice can contain two distinct “puncture” Boolean blocks that (a) each faithfully and order-separatingly represent the same countable decreasing clopen neighbourhood basis at a point, and (b) identify that shared boundary data between the two blocks. Concretely: if $(U_n)_{n < \omega}$ is a decreasing sequence of clopen sets in a Cantor-space-like block converging to a single point i , and this same countable boundary sequence is glued identically into a second block converging instead to a different point $j \neq i$, then the countable meet $\bigwedge_n U_n$ evaluates to $\{i\}$ in the first block’s intrinsic order and to (something containing) $\{j\} \neq \{i\}$ in the second — an outright contradiction, since the meet of a fixed countable family is an intrinsic feature of the ambient σ -complete lattice and cannot depend on which block computes it.*

Proof sketch. The meet of a decreasing countable family in a σ -complete lattice is computed once, in the ambient structure, via complementation and the countable-join closure; it cannot simultaneously equal two different singleton-supported values as seen from two different faithful embeddings, since both embeddings realize the *same* ambient meet. $\square$

What this buys. This rules out the most naive way of trying to build a witness by gluing together countably many copies of a Cantor-space-like Boolean piece along a shared boundary: the gluing itself is already inconsistent with latticehood, independently of centrality (independently, that is, of whether the shared piece sits in the “classical center” of the structure or not). It is one of the three obstruction results whose combined weight is discussed in Section 5.

Proposition 3.7 (The countably-generated route is fully closed). *Combining Theorem 3.3 with a standard fact about standard Borel spaces (the Lusin–Souslin injection theorem: if countably many countably-generated quotient algebras of a standard Borel space jointly separate its points, the algebra they generate is already the full Borel σ -algebra), one obtains: no countably generated, faithfully and separatingly embedded boundary structure can support the missing σ -state either — the same triviality forced by Theorem 3.3 recurs however the countably-generated pieces are combined, provided they remain countably many.*

What this buys. It closes off, completely, the entire family of constructions built from only countably much combinatorial data. Any surviving construction has to be genuinely, unavoidably uncountably-generated — this sharpens “the difficulty is infinitary” from a vague slogan into a theorem.

4 The one existing witness: a proved ZFC theorem

We now state, precisely, the one witness that has actually been constructed and proved. It settles the orthomodular-*poset* form of the question completely; it is the reason the poset/lattice distinction is the entire remaining content.

Theorem 4.1 (A σ -essential contextual state exists, in ZFC). *There is a concrete σ -complete orthomodular poset L (built honestly from subsets of a set Ω of cardinality $\aleph_1$, the first uncountable cardinal) carrying a finite pattern s_0 that is a σ -essential contextual state, in the sense of Definition 1.7. The proof uses only the ordinary axioms of ZFC (in particular, the axiom of choice in its usual strength — the proof cites the ultrafilter/prime-ideal principle plus the standard fact that every filter extends to an ultrafilter, both consequences of full choice as normally assumed) — no large cardinal or independence hypothesis of any kind is invoked or needed.*

Proof sketch, at the conceptual level (full detail is a substantial independent development, sketched here in outline). The construction has three moving parts, deliberately kept from interacting except through countable “noise”:

- (a) **An Ulam matrix.** On a set M of size $\aleph_1$ with a well-ordering whose every initial segment is countable, one builds (using choice, one injection per ordinal) a doubly-indexed family of subsets $\{C_{\alpha,n} : \alpha < \omega_1, n < \omega\}$ such that: each “row” $\{C_{\alpha,n}\}_{n < \omega}$ is pairwise disjoint and covers all of M except a countable set; and each “column” $\{C_{\alpha,n}\}_{\alpha < \omega_1}$ (fixed n) is pairwise disjoint across different rows. This is a classical device going back to Ulam’s work on measurable cardinals, repurposed here for a different end.
- (b) **A rigidity theorem.** Enlarge the carrier to $\Omega = M \times \{1, 2, 3, 4\}$ (four “fibers” over each point of M), and let L be generated by all singletons, by the “cells” $D_{\alpha,n} := C_{\alpha,n} \times \{1, 2, 3, 4\}$, and by three further distinguished sets A, B, C described in part (c). One proves: every σ -additive two-valued state on L must be a Dirac (point) state. The argument is a clean pigeonhole: if a state kills every singleton (hence, by σ -additivity, every countable set), then in each row exactly one cell must carry value 1; but only countably many cells exist per row,

while there are uncountably many rows, so two different rows are forced (by the pigeonhole principle applied to an uncountable-into-countable map) to select the *same* column index; the corresponding two cells are disjoint by the Ulam matrix's column-disjointness property, so additivity would force their union to have value 2 — impossible for a $\{0, 1\}$ -valued state. Hence no non-Dirac σ -state can exist at all: rigidity.

- (c) **A finite pattern realized by an ultrafilter's “vote,” but by no point.** The three sets $A = M \times \{1, 2\}$, $B = M \times \{1, 3\}$, $C = M \times \{2, 3\}$ have every pairwise intersection nonempty (e.g. $A \cap B = M \times \{1\}$) but empty triple intersection ($A \cap B \cap C = \emptyset$) — the value-1 family of the intended pattern fails the finite intersection property at the smallest possible stage, three, exactly matching Theorem 3.5's necessary condition. One then fixes, using choice, a non-principal ultrafilter on M and uses it to define a finitely additive two-valued state m on all of L that assigns value 1 to each of A, B, C simultaneously (a short parity computation shows the three assignments are jointly consistent even though no single point can witness all three at once). This m certifies finite coherence: the pattern $s_0 := m$ restricted to the finite piece generated by A, B, C is a legitimate two-valued state on a finite sub-poset, extendable (by m itself) to a full finitely additive state on all of L .

Putting the pieces together: s_0 is finitely coherent (by the state m), but by rigidity (part (b)) the *only* candidates for a σ -additive extension are Dirac states, and no point $\omega \in \Omega$ lies in $A \cap B \cap C = \emptyset$ — so no Dirac state extends s_0 either. Hence s_0 is a σ -essential contextual state, proving the theorem. $\square$

Why the carrier is not a lattice. A key structural fact about this witness, worth stating on its own, is that the carrier L constructed above is provably *not* a lattice.

Proposition 4.2 (The witness is an orthomodular poset, not a lattice). *In the carrier L of Theorem 4.1, the two sets A and B have no greatest lower bound inside L : every member of L contained in $A \cap B = M \times \{1\}$ is a countable subset of that set, and among the countable subsets of an uncountable set there is no largest one (one can always adjoin one more missing point and remain inside L , since L contains every countable set). So the meet $A \wedge B$ simply fails to exist in L : L is a genuine orthomodular poset, not an orthomodular lattice.*

What this buys, and why it is the crux of the whole document. It is precisely this fact — that the one proved witness lives on a *poset*, not a *lattice* — that keeps Conjecture 2.1 open. The theorem above resolves the poset form of the question completely and unconditionally; it says nothing, one way or the other, about the lattice form, because the object it constructs is not a lattice. Every later section is, in one way or another, about this exact gap.

5 The external boundary result: measure extension on operator algebras

A recent and closely related development sits just outside this programme's own object of study, in a genuinely different (though related) mathematical world — the projection lattices of certain operator algebras — and it is important to state precisely what it does and does not tell us.

External Theorem 5.1 (Escolano–Peralta–Villena, 2025). *Let J be a JBW^* -algebra (a certain class of complex Jordan Banach algebras that generalize von Neumann algebras and includes the self-adjoint operator algebras relevant to quantum mechanics), and let $P(J)$ denote its lattice of*

projections. Suppose J has no direct summand of type I_2 (type I_2 is the smallest genuinely non-distributive building block that can occur in this classification — essentially, a spin factor, the operator-algebraic analogue of the lattice MO_2 from Example 1.2). Then:

- (i) every bounded finitely additive real- or complex-valued measure on $P(J)$ extends to a bounded linear functional on J ;
- (ii) more generally (via Hahn–Banach), every bounded finitely additive measure on $P(J)$ valued in an arbitrary Banach space extends to a bounded linear operator on J .

Sharpness (the dual result). If J does have a type I_2 summand, this fails: there exists a positive, bounded, finitely additive measure on $P(J)$ that does not extend to a bounded linear functional. So the no- I_2 hypothesis is exactly sharp, not merely sufficient.

The mechanism, at a conceptual level. The proof of the positive direction (i) does not proceed via any infinitary or set-theoretic device. It establishes that, away from type I_2 , every bounded finitely additive measure on the projection lattice is automatically *uniformly continuous* — nearby projections receive close measure values — and uniform continuity is then upgraded to full linear extendability by standard functional-analytic machinery. The uniform-continuity step itself rests on three purely geometric/algebraic operations available on projections once type I_2 is excluded: *halving* a projection into two smaller, mutually equivalent orthogonal pieces; *small symmetric motions* that exchange two nearby projections while moving everything else only a controlled amount; and *isoclinic interpolation*, inserting a third projection at a controlled angle to two given ones so as to bridge between them in a finite number of algebraic steps. None of this uses anything about uncountable cardinals, ultrafilters, or countable additivity in the sense of this document’s Definition 1.5 — it is, in the relevant technical sense, an *elementary* (finite, constructive) mechanism.

Remark 5.2 (What this does and does not tell us for the lattice problem). Theorem 5.1 is genuinely useful, and genuinely limited, in the following precise ways.

What it does. It *narrows the target*: if a witness for the analogous phenomenon in the JBW^* -projection-lattice world exists at all, it must live in the type- I_2 , “symmetry-poor” regime — everywhere else, the theorem forbids it outright. This gives, for the first time, a precise structural profile for where such a phenomenon-of-this-general-shape can occur in that adjacent world.

What it does not do. Three separate limitations, each load-bearing:

- *Different carriers.* The theorem is about JBW^* projection lattices. The concrete σ -complete orthomodular lattices and posets that are the subject of this entire document (Definition 1.4) are not, in general, JBW^* projection lattices — they need not carry any compatible norm or Jordan algebra structure at all. The theorem simply does not apply to them directly, in either direction: it neither forbids nor supplies a witness on the concrete carriers of interest here.
- *The mechanism does not match the obstruction needed here.* Even restricted to its own setting, the dual (sharpness) direction of Theorem 5.1 — the existence of a non-extendable measure on a type- I_2 summand — arises from an elementary, finite piece of linear algebra (a rank-counting contradiction inside a small spin factor: a measure assigning value $\frac{1}{2}$ to rank-one projections forces an arithmetic clash, $\frac{1}{3} \neq \frac{1}{2}$, in a specific three-dimensional computation). This is categorically *not* the kind of obstruction the lattice conjecture (Conjecture 2.1) needs: that conjecture, if false, would need a genuinely *infinitary*, ω_1 -scale combinatorial phenomenon (exactly the kind built in Theorem 4.1), not a finite linear-algebra clash. The two mechanisms live at different scales entirely.

- *It narrows a location, not a proof.* Even taking the “symmetry-poor, type- I_2 -rich” heuristic at face value as a suggestion for *where*, in the concrete-lattice world, a witness might have to live (were one to exist), it supplies no argument for *whether* one exists there. It is a candidate-location filter, not a partial proof.

Summary of this section, in one sentence. Theorem 5.1 is a genuine, recent, sharp external result that narrows where an analogous phenomenon could live in an adjacent mathematical world, but it neither applies to, nor resolves, nor even meaningfully constrains a proof strategy for, the concrete-lattice conjecture that is the subject of this document; it is correctly read as *narrowing a target*, not as *a step toward a solution*.

6 Where the difficulty actually is: the honest crux

We now assemble what has been built up into a single, plain-language account of exactly what makes Conjecture 2.1 hard, and why every finite technique available provably cannot reach it.

If a witness for the *lattice* form of the conjecture exists, it must satisfy four requirements simultaneously, each individually forced by the results of Sections 3–4:

- (i) **Genuinely uncountably-generated.** By Theorem 3.3 and Proposition 3.7, no witness can be built from — or faithfully, separably embedded into — countably much combinatorial data. The one existing (poset-level) witness needs a carrier of size $\aleph_1$; nothing smaller is known to suffice, and the countably-generated route is entirely closed.
- (ii) **Living in a “symmetry-poor” regime.** Insofar as the external boundary result (Theorem 5.1, understood only as a heuristic pointer in the different-but-related operator-algebra world) is any guide at all, the relevant phenomena cluster where the local algebraic symmetry is minimal — the type- I_2 -like corner, not a richly homogeneous one.
- (iii) **Carrying a genuinely contextual σ -additive-seeming state.** By Theorem 3.5, the finite pattern in question must fail the finite intersection property already at some finite stage; by Definition 1.7, it must nonetheless be finitely coherent — i.e. it is not merely inconsistent nonsense, it is a real finite fragment of some finitely-additive global assignment, which is what makes its failure to extend σ -additively meaningful rather than vacuous.
- (iv) **Surviving lattice closure.** This is the hard core, and it deserves its own explanation.

6.1 Why requirement (iv) is the genuine obstacle

The one proved witness (Theorem 4.1) achieves (i)–(iii) with room to spare. What it cannot achieve — and what Proposition 4.2 shows it provably fails to achieve — is (iv): its carrier is not a lattice. The two events A and B have no greatest common refinement in the structure at all. This is not an accident of one particular construction; it recurs across every construction this programme has attempted (the “countable-meet obstruction” of Proposition 3.6 is a second, independent instance of the same phenomenon, arising from a completely different building strategy). In every case examined, the mechanism that makes the finite pattern genuinely essential (no single point can simultaneously satisfy every constraint) is the *same* mechanism that, when one tries to force the structure to have all its meets and joins after all, either destroys the essential pattern (some meet, once it is forced to exist, ends up realizing the pattern after all, at a single point) or is simply unavailable to compute (the required meet provably does not exist, as in Proposition 4.2).

Concretely: requiring L to be a lattice means requiring, for *every* pair of elements, that a specific new element — their meet — be manufactured and adjoined to the structure if it is not already present, in a way that is consistent with everything else (associativity, the orthomodular law, all prior additivity constraints). Doing this for *countably many* pairs at once, in a coherent, simultaneous way, is already delicate; doing it for the *uncountably many* pairs that necessarily arise once the carrier is forced to be uncountable (requirement (i) above) is a fundamentally different order of combinatorial difficulty. It is, in the precise technical sense used throughout this document, a question about ω_1 -level combinatorics: whether a coherent system of forced meets and joins can be built across an uncountable index set without collapsing the essential pattern into something a single point can witness after all.

Remark 6.1 (Why finite methods cannot reach this). Every finite computational or combinatorial technique this programme has applied — building explicit finite orthomodular lattices of larger and larger size, checking by exhaustive case analysis whether some assembly of finitely many compatible pieces can carry an essential pattern, searching for finite “repair” operations that patch a near-miss into a genuine lattice — has, without exception, produced only *tame* objects (Proposition 3.6: every finite orthomodular poset is automatically tame, for the trivial reason that there is no infinite limit to fail to respect). This is not a series of unlucky misses; it is forced by the definitions. **No finite construction can ever exhibit requirement (iv) failing at the uncountable scale that requirement (i) demands**, because finiteness makes σ -additivity automatic. The genuinely infinitary question — can lattice closure be achieved, across an uncountable index set, without ever realizing the pattern at a single point — is simply not a question that any finite search, however large, can settle either way. This is why the problem has resisted direct construction: the tools available for exploring finite and countable structures are, provably, the wrong tools for this particular question.

6.2 The current assessment

Remark 6.2 (Where the evidence leans — an assessment, not a theorem). Based on everything established so far, the honest assessment is that the evidence leans toward Conjecture 2.1 being **true** (no lattice witness exists), though this is emphatically an assessment, not a proof, and it rests on the following basis: (a) the two-cell countable-meet obstruction (Proposition 3.6), which rules out the most natural gluing construction independently of any centrality assumption; (b) the countably-generated boundary extension theorem (Theorem 3.3) together with its standard-Borel strengthening (Proposition 3.7), which together close off every construction built from only countably much data; (c) a normality trichotomy (not separately stated above as a numbered result, but of the same character: attempts to build an uncountable “hub” structure with a genuinely non-classical shared boundary, while preserving σ -state order-separation throughout, have in every case examined either forced the shared boundary back into the classical center of the structure or destroyed order-separation outright); and (d) every attempt so far to identify a genuinely lattice-theoretic mechanism that would let an essential pattern survive full lattice closure has, on inspection, either turned out to require exactly the Boolean/distributive structure the conjecture is about ruling out, or has been shown, on careful examination, not to supply the needed mechanism at all (see Section 7 for two such falsified attempts, examined and rejected in detail). If Conjecture 2.1 is eventually proved, it would close this entire line of investigation positively: it would say that requiring genuine lattice structure is, by itself, already enough to force every finitely coherent finite pattern to extend σ -additively — a clean and strong structural theorem in its own right, independent of any of the constructions above.

7 The two open directions, precisely

Two distinct, precisely statable mathematical questions remain open, and resolving either one — in either direction — would constitute genuine progress. Both are naturally read with a two-sided framing: either direction of a proof (the obstruction generalizes / the escape succeeds) tells you something, either a theorem or a design constraint on any future witness attempt.

7.1 Route (a): does the countable-meet obstruction generalize?

Conjecture 7.1 (Generalized countable-meet obstruction). *Does the two-cell obstruction of Proposition 3.6 generalize from the specific Cantor-space-like gluing considered there to every faithful, order-separating, countably-generated “hub” structure shared between two or more blocks of a concrete σ -complete orthomodular lattice — i.e., is every attempt to identify a shared countably-generated separating boundary between distinct blocks of a lattice necessarily inconsistent, by the same countable-meet mechanism, regardless of the specific combinatorial shape of the gluing?*

If proved: this would be a genuine step toward Conjecture 2.1, by eliminating an entire broad class of candidate constructions (all countably-generated-boundary gluings) at once, rather than one construction at a time. *If it fails* (some faithful countably-generated hub structure *does* survive lattice closure without the countable-meet contradiction firing): the failure would itself be a genuine discovery, supplying — for the first time — a concrete design template for a lattice-level witness attempt, built around whatever specific feature let it escape the obstruction.

7.2 Route (b): the distributed nonseparating quotient route

Conjecture 7.2 (Distributed nonseparating quotient route). *Is there a concrete σ -complete orthomodular lattice built from an uncountably-generated boundary structure in which the shared boundary data is not required to order-separate points across the whole structure — i.e., one that evades Theorem 3.3’s hypothesis (which requires order-separation) by working instead with a distributed family of nonseparating quotient copies of a shared structure, glued so that no single global σ -state can be pieced together across the whole family, while every finite and countable sub-piece remains individually consistent?*

If proved (such a structure can be built as a genuine lattice): this would refute Conjecture 2.1 outright and supply the first lattice-level witness, closing the poset/lattice gap from the other side. *If it can be shown impossible* (every attempt at such a distributed nonseparating gluing, when forced to be a genuine lattice, either collapses to order-separation after all or fails to remain σ -complete): this would be the second and complementary half of a full proof of Conjecture 2.1, alongside a positive resolution of Route (a).

Remark 7.3 (The two routes as complementary halves). Routes (a) and (b) are naturally read together: (a) attacks the *countably-generated* case in full generality; (b) attacks the genuinely *uncountably-generated, nonseparating* case that (a) alone does not cover. A full proof of Conjecture 2.1 would most naturally proceed by closing both simultaneously (every countably generated hub fails by (a)’s mechanism; every uncountably-generated attempt fails by some extension of the ideas behind (b)’s impossibility direction). A witness, if one exists at all, must in particular refute route (b) constructively while consistently already being ruled out by nothing resembling route (a)’s mechanism.

8 What is heuristic, not yet mathematics

Honesty about the boundary between established mathematics and intuition requires flagging, explicitly, several ideas that were seriously explored and did *not* survive scrutiny, so that a reader does not later mistake a dead idea for a live one.

Falsified 8.1 (A “dimensional-equivalence” relation as a proxy for shared-boundary agreement). One natural-seeming heuristic, motivated loosely by the external boundary result of Section 4, conjectured that a specific order-theoretic relation between atoms of an orthomodular lattice — informally, “these two atoms are related by a dimension-preserving symmetry of the structure,” the lattice-theoretic analogue of Murray–von Neumann equivalence of projections in an operator algebra — might serve as a faithful proxy for the entirely different relation of “these two events are forced to agree under every state that assigns them both value 1 on a shared boundary” (the relation actually doing the work in the assembly obstructions of Section 3). **This was tested directly and falsified.** On the pentagon (Example 1.3) and on finite homogeneous orthomodular lattices generally, the dimensional-equivalence relation turns out to be *completely non-selective*: every atom is related to every other atom, without exception, purely as a structural consequence of finite homogeneity — any two atoms in such a lattice can always be connected by a chain of complementary pairs. A relation that relates *everything* to everything cannot possibly serve as a faithful proxy for a relation (shared-boundary agreement) that is supposed to be selective — picking out specific structured pairs and not others. The two relations are, moreover, of fundamentally different type: one is rank-sensitive (a property of how projections compare in size or dimension), the other is rank-blind (a property of which events happen to agree under a shared state, regardless of any notion of size). No finite example can rescue this identification, because the non-selectivity is a structural feature of finite homogeneous lattices generally, not an accident of the particular examples tested. This heuristic is dead; it should not recur in future attempts under a new name.

Falsified 8.2 (A parity/winding invariant tracking a would-be obstruction). A related heuristic conjectured that some integer- or parity-valued “winding” invariant, attached to how a proposed lattice-completion process forces meets and joins around a cycle of incompatible events (as in the pentagon, Example 1.3), might detect exactly when a forced completion is compatible with an essential pattern surviving versus when it is not — the hoped-for mechanism for making requirement (iv) of Section 5 mathematically tractable. This heuristic was built directly on top of the dimensional-equivalence proxy just falsified (Falsified Claim 8.1), and falls with it: once the underlying relation it was meant to track is shown to carry no selective information, the invariant built from it carries none either. Moreover, the two natural test objects available — the finite pentagon (which, being finite, is automatically tame by Proposition 3.6, so it can only ever *illustrate*, never genuinely test, a forced-closure phenomenon) and the one proved witness of Theorem 4.1 (which, by Proposition 4.2, is provably *not* a lattice, so it has no forced meets to have a winding invariant *about* in the first place) — are, for structural reasons, incapable of ever exhibiting the phenomenon the invariant was designed to detect, regardless of how the invariant itself is defined. This is a second, independent reason the idea does not survive: even setting aside Claim 8.1, there is currently no finite or countable object on which the hoped-for invariant could ever be observed to do anything at all.

Heuristic 8.3 (A legitimate, still-unexplored forwarding lead). One genuine mathematical object, distinct from the falsified ideas above and not yet investigated by this programme in any depth, is worth naming explicitly as a place a parity-type idea *might* legitimately live: the study, in the published literature on orthomodular structures closed under a symmetric-difference operation, of $\mathbb{Z}_2$ -valued states — states valued in the two-element group rather than in $\{0, 1\}$ read as real

numbers, studied for their own sake as an algebraic invariant of such structures. This is a real, previously studied class of objects in the literature (distinct in kind from the ad hoc winding invariant of Falsified Claim 8.2, which was invented for this programme’s purposes and does not appear in the literature under any name). Whether these $\mathbb{Z}_2$ -valued states have anything to do with the essential-witness question of this document is entirely open and unexplored; it is flagged here only as a legitimate place to look next, not as a result, a partial result, or even a properly formulated conjecture yet.

Closing remark

The problem, in full, is this. A witness to genuinely contextual, σ -inextendable probability data has been constructed and proved, unconditionally, in ZFC — but only on a structure one order-theoretic notch weaker than the one the field ultimately cares about (an orthomodular poset rather than a genuine lattice). Whether the extra requirement of latticehood is, by itself, already enough to force every such contextual pattern to dissolve is the open question. Three independent obstruction results and one recent external theorem in an adjacent world all point the same direction, toward the conjecture being true, but the actual reason the obstruction is hard to prove outright is structural and precise: it is a genuinely infinitary, ω_1 -combinatorial question about whether an uncountable family of forced lattice meets and joins can be assembled coherently without ever collapsing the pattern to a single point, and no finite method — however large or clever — is even in principle capable of reaching a question of that shape. Two precise routes remain open, stated in Section 6; either one’s resolution, in either direction, would be genuine progress on a question that currently sits exactly at the boundary between what is known and what is not.